WORKED SOLUTION

STEP II 2004, Question 5

Pure • Integral Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q5

The key idea

The unknown function sits inside an integral, which looks frightening until you expand $\sin(x+t)$ with the compound-angle formula. Then t comes outside the integral, and what is left inside are two fixed *numbers*. So f must be t plus a combination of $\sin t$ and $\cos t$, and the whole problem reduces to finding two constants from two linear equations. The two integrals in the first part are exactly the ones that equation needs.

Solution

The two integrals

Integrating by parts with $u = x$ and $\mathrm{d}v = \sin x\,\mathrm{d}x$:

$$\int_0^\pi x\sin x\,\mathrm{d}x = \Big[-x\cos x\Big]_0^\pi + \int_0^\pi \cos x\,\mathrm{d}x = (-\pi\cos\pi - 0) + \Big[\sin x\Big]_0^\pi = \pi.$$

Similarly, with $u = x$ and $\mathrm{d}v = \cos x\,\mathrm{d}x$:

$$\int_0^\pi x\cos x\,\mathrm{d}x = \Big[x\sin x\Big]_0^\pi - \int_0^\pi \sin x\,\mathrm{d}x = 0 - \Big[-\cos x\Big]_0^\pi = -(1+1) = -2.$$

We will also need three standard integrals over $[0,\pi]$:

$$\int_0^\pi \sin^2 x\,\mathrm{d}x = \int_0^\pi \cos^2 x\,\mathrm{d}x = \frac{\pi}{2}, \qquad \int_0^\pi \sin x\cos x\,\mathrm{d}x = 0.$$

The first two follow from $\sin^2 x = \frac{1}{2}(1-\cos 2x)$ and $\cos^2 x = \frac{1}{2}(1+\cos 2x)$, since $\cos 2x$ integrates to zero over a whole period; the third is $\frac{1}{2}\int_0^\pi \sin 2x\,\mathrm{d}x = 0$ for the same reason.

The form of f

Expand $\sin(x+t) = \sin x \cos t + \cos x \sin t$. The integration is over x, so $\sin t$ and $\cos t$ are constants and come out of the integral:

$$\begin{aligned} \mathrm{f}(t) &= t + \int_0^\pi \mathrm{f}(x)(\cos x \sin t + \sin x \cos t)\,\mathrm{d}x \\ &= t + \sin t \int_0^\pi \mathrm{f}(x)\cos x\,\mathrm{d}x + \cos t \int_0^\pi \mathrm{f}(x)\sin x\,\mathrm{d}x \\ &= t + A\sin t + B\cos t, \end{aligned}$$

where A and B are the two given integrals. Since they don't depend on t, they are genuine constants. This is the required form.

Finding A and B

? **Ask yourself** We don't know f, so how can we work out A and B?

We do know its *shape*: $\mathrm{f}(x) = x + A\sin x + B\cos x$. Putting this back into $(*)$ turns the integral into something we can evaluate in terms of A and B, and matching the two sides gives equations for them.

Substitute into the right-hand side of $(*)$, using the expansion of $\sin(x+t)$ from above:

$$t + \sin t \int_0^\pi (x + A\sin x + B\cos x)\cos x\,\mathrm{d}x + \cos t \int_0^\pi (x + A\sin x + B\cos x)\sin x\,\mathrm{d}x.$$

Using the integrals found earlier,

$$\begin{aligned} \int_0^\pi (x + A\sin x + B\cos x)\cos x\,\mathrm{d}x &= -2 + A\cdot 0 + B\cdot\frac{\pi}{2} = -2 + \frac{\pi}{2}B, \\ \int_0^\pi (x + A\sin x + B\cos x)\sin x\,\mathrm{d}x &= \pi + A\cdot\frac{\pi}{2} + B\cdot 0 = \pi + \frac{\pi}{2}A. \end{aligned}$$

So $(*)$ says that, for every t,

$$t + A\sin t + B\cos t = t + \left(-2 + \frac{\pi}{2}B\right)\sin t + \left(\pi + \frac{\pi}{2}A\right)\cos t.$$

Equating coefficients of $\sin t$ and $\cos t$ (justified because the identity holds for all t: putting $t = \frac{\pi}{2}$ and then $t = 0$ isolates each coefficient) gives

$$A = -2 + \frac{\pi}{2}B, \qquad B = \pi + \frac{\pi}{2}A.$$

Substituting the second into the first,

$$A = -2 + \frac{\pi^2}{2} + \frac{\pi^2}{4}A \quad\Longrightarrow\quad A\left(1 - \frac{\pi^2}{4}\right) = \frac{\pi^2}{2} - 2, \quad \text{i.e.} \quad A\cdot\frac{4-\pi^2}{4} = -\frac{4-\pi^2}{2}.$$

Since $\pi^2 \neq 4$, we may divide by $4 - \pi^2$, giving $A = -2$. Then $B = \pi + \frac{\pi}{2}(-2) = 0$. Therefore

$$A = -2, \qquad B = 0, \qquad \text{so} \qquad \mathrm{f}(t) = t - 2\sin t.$$

Checks

Check the answer against the definitions of A and B directly. With $\mathrm{f}(x) = x - 2\sin x$,

$$\int_0^{\pi} \mathrm{f}(x)\cos x\,\mathrm{d}x = -2 - 2\cdot 0 = -2 = A, \qquad \int_0^{\pi} \mathrm{f}(x)\sin x\,\mathrm{d}x = \pi - 2\cdot\frac{\pi}{2} = 0 = B.$$

Numerically, Simpson's rule gives $\int_0^{\pi} x\sin x\,\mathrm{d}x \approx 3.14159$ and $\int_0^{\pi} x\cos x\,\mathrm{d}x \approx -2.00000$. We also evaluated the right-hand side of $(*)$ with $\mathrm{f}(t) = t - 2\sin t$ at fifty random values of t, and it matched $\mathrm{f}(t)$ to within 10^{-9} every time.

Common slips

- Forgetting the minus sign in $\int_0^{\pi} x\cos x\,\mathrm{d}x$: the boundary term vanishes and $-\int_0^{\pi}\sin x\,\mathrm{d}x = -2$.
- Mixing up which integral multiplies $\sin t$. The $\cos x\sin t$ part of $\sin(x+t)$ gives $A\sin t$ with $A = \int \mathrm{f}(x)\cos x\,\mathrm{d}x$, not $\int \mathrm{f}(x)\sin x\,\mathrm{d}x$.
- Dividing by $1 - \frac{\pi^2}{4}$ without noting that it is non-zero.

Beyond STEP

Equation $(*)$ is a *Fredholm integral equation of the second kind*. Its kernel $\sin(x+t) = \sin x\cos t + \cos x\sin t$ is a finite sum of products of a function of x and a function of t (a "degenerate" or "separable" kernel), and that is exactly why the problem collapses to a 2×2 linear system. The condition $\pi^2 \neq 4$ is the statement that this system has a non-zero determinant; if it were zero there could be no solution or infinitely many, which is the integral-equation version of a singular matrix. For the linear algebra behind this, see Strang, *Introduction to Linear Algebra*.

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